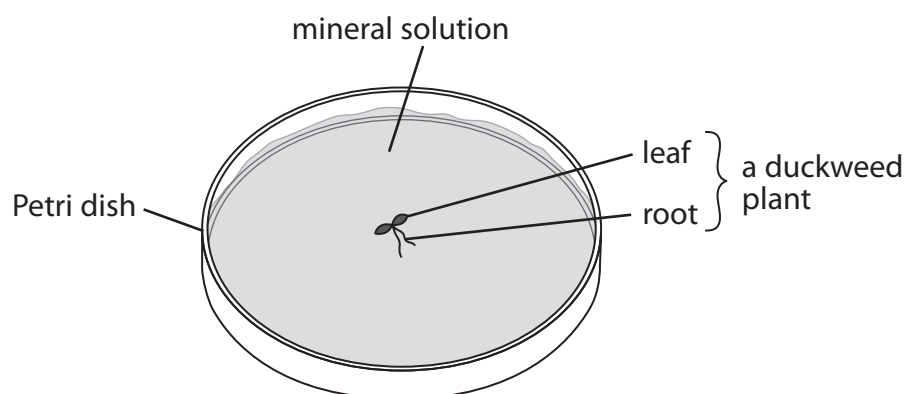


- 2 Duckweed is a water plant. These plants multiply by producing buds that become separate plants.

The effect of minerals on the growth of duckweed plants was investigated.

A single duckweed plant was placed in each of seven Petri dishes containing 20 cm^3 of a mineral solution, as shown in the diagram.



Each mineral solution was missing one of the minerals that plants need to grow.

The number of duckweed plants in each dish was counted after 14 days.

- (a) (i) Describe the contents of two more Petri dishes that could be controls for this investigation.

(2)

Dish 1

Dish 2

- (ii) In this investigation, temperature and pH are variables that need to be controlled.

Describe how these variables could be controlled in this investigation.

(2)

Temperature

pH



(b) The table shows the results of this investigation.

Mineral missing from the solution	Number of duckweed plants after 14 days
calcium	10
iron	35
potassium	24
magnesium	12
nitrate	5
phosphate	14
sulfate	27

(i) Plot a suitable graph to show the results of this investigation.

(4)



- (ii) Explain the difference in duckweed growth in the mineral solution missing nitrate and the mineral solution missing phosphate.

Use the information in the table to support your answer.

(3)

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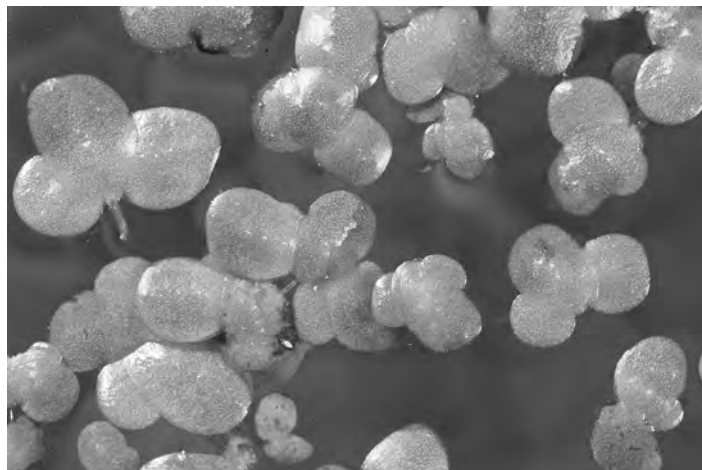
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P 6 6 1 5 5 A 0 9 2 0

(c) The photograph shows several duckweed plants in one of the Petri dishes.



Magnification $\times 2$

(Source: © BOB GIBBONS / SCIENCE PHOTO LIBRARY)

- (i) Suggest why counting plants will give an inaccurate result for the growth of duckweed after 14 days.

Use the information in the photograph to support your answer.

(2)

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- (ii) Explain how this investigation could be modified to provide an accurate measure of the growth rate of duckweed.

(4)

(Total for Question 2 = 17 marks)

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